

- **Master-Slave** architecture in Jenkins, now often referred to as **Controller-Agent** in modern Jenkins versions, is a way to distribute tasks across multiple machines to improve scalability, efficiency, and fault tolerance.
- The **master-slave** concept in Jenkins refers to a distributed architecture model that enhances scalability, efficiency, and fault tolerance in Jenkins environments.

Master (or **Controller** in newer versions) is the central node in Jenkins. It is responsible for the orchestration of Jenkins jobs, including:

- Scheduling builds
- Monitoring the status of builds
- Storing configuration, job data, and logs
- Managing the Jenkins web interface

Slave (or **Agent**) is a remote machine that performs the actual build execution. A slave can be a physical or virtual machine, and it runs the build tasks assigned by the master. Slaves help distribute the computational load across multiple machines.

Launched master and slave instances

Instances (2) [Info](#)

Last updated 9 minutes ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

All states ▼

running X

Clear filters

< 1 >

☐	Name	▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone ▼	Public IP
☐	master		i-01c055e38b6ad9c50	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1d	ec2-3
☐	slave		i-0750e989f9202fd87	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1d	ec2-1

Select an instance

▼

Logged into master node

Dashboard > Manage Jenkins

 Check File Fingerprint

 Manage Jenkins

 My Views

Build Queue

No builds in the queue.

Build Executor Status

 Built-In Node

0/2

 slave-1

0/2

System Configuration



System

Configure global settings and paths.



Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.



Tools

Configure tools, their locations and automatic installers.



Clouds

Add, remove, and configure cloud instances to provision agents on-demand.



Plugins

Add, remove, disable or enable plugins that can extend the functionality of Jenkins.



Appearance

Configure the look and feel of Jenkins

Security



Security

Secure Jenkins; define who is allowed to access/use the system.



Users

Create/delete/modify users that can log in to this Jenkins.



Credentials

Configure credentials



Credential Providers

Configure the credential providers and types

Logged into slave node

```
ubuntu@ip-172-31-91-190:~$ sudo systemctl start jenkins
ubuntu@ip-172-31-91-190:~$ sudo systemctl status jenkins
• jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/lib/systemd/system/jenkins.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2024-12-07 13:50:30 UTC; 52s ago
     Main PID: 5094 (java)
        Tasks: 41 (limit: 1130)
       Memory: 269.1M
          CPU: 15.528s
      CGroup: /system.slice/jenkins.service
              └─5094 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPor

Dec 07 13:50:23 ip-172-31-91-190 jenkins[5094]: e58bc305f8164ea4abafcd61747bf156
Dec 07 13:50:23 ip-172-31-91-190 jenkins[5094]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Dec 07 13:50:23 ip-172-31-91-190 jenkins[5094]: *****
Dec 07 13:50:23 ip-172-31-91-190 jenkins[5094]: *****
Dec 07 13:50:23 ip-172-31-91-190 jenkins[5094]: *****
Dec 07 13:50:30 ip-172-31-91-190 jenkins[5094]: 2024-12-07 13:50:30.324+0000 [id=31] INFO jenkins.InitReactorRunner$1#on
Dec 07 13:50:30 ip-172-31-91-190 jenkins[5094]: 2024-12-07 13:50:30.348+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onR
Dec 07 13:50:30 ip-172-31-91-190 systemd[1]: Started Jenkins Continuous Integration Server.
Dec 07 13:50:30 ip-172-31-91-190 jenkins[5094]: 2024-12-07 13:50:30.604+0000 [id=46] INFO h.m.DownloadService$Downloadab
Dec 07 13:50:30 ip-172-31-91-190 jenkins[5094]: 2024-12-07 13:50:30.604+0000 [id=46] INFO hudson.util.Retrier#start: Per

ubuntu@ip-172-31-91-190:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-91-190:~$ mkdir jenkins
ubuntu@ip-172-31-91-190:~$ cd jenkins
ubuntu@ip-172-31-91-190:~/jenkins$ pwd
/home/ubuntu/jenkins
ubuntu@ip-172-31-91-190:~/jenkins$
```

Master node-mange jenkins-configured details about slave node

Remote root directory ?

/home/ubuntu/jenkins

Labels ?

dev

Usage ?

Use this node as much as possible



Launch method ?

Launch agents via SSH



Host ?

18.208.156.109

Credentials ?

ubuntu (gmailcredential123)



Save

Launch method ?

Launch agents via SSH

▼

Host ?

18.208.156.109

Credentials ?

ubuntu (gmailcredential123)

▼

+ Add

Host Key Verification Strategy ?

Manually trusted key Verification Strategy

▼

☐ Require manual verification of initial connection ?

Advanced ▼  Edited

Availability ?

Keep this agent online as much as possible

▼

Save

Launched agent

Dashboard > Manage Jenkins > Nodes > slave-1



Status



Delete Agent



Configure



Build History



Load Statistics



Script Console



Log



System Information



Disconnect

Agent slave-1

Edit description

this is slave node

Monitoring Data ▾

Labels

dev

Projects tied to slave-1

None

Build Executor Status

0/2 ▾

Slave node log

Dashboard > Nodes > slave-1 > Log

```
- .
Checking Java version in the PATH
openjdk version "17.0.13" 2024-10-15
OpenJDK Runtime Environment (build 17.0.13+11-Ubuntu-2ubuntu122.04)
OpenJDK 64-Bit Server VM (build 17.0.13+11-Ubuntu-2ubuntu122.04, mixed mode, sharing)
[12/07/24 14:05:20] [SSH] Checking java version of /home/ubuntu/jenkins/jdk/bin/java
Couldn't figure out the Java version of /home/ubuntu/jenkins/jdk/bin/java
bash: line 1: /home/ubuntu/jenkins/jdk/bin/java: No such file or directory

[12/07/24 14:05:20] [SSH] Checking java version of java
[12/07/24 14:05:20] [SSH] java -version returned 17.0.13.
[12/07/24 14:05:20] [SSH] Starting sftp client.
[12/07/24 14:05:20] [SSH] Copying latest remoting.jar...
[12/07/24 14:05:20] [SSH] Copied 1,393,083 bytes.
Expanded the channel window size to 4MB
[12/07/24 14:05:20] [SSH] Starting agent process: cd "/home/ubuntu/jenkins" && java -jar remoting.jar -workDir /home/ubuntu/jenkins -jar-cache /home/ubuntu/jenkins/remoting/jarCache
Dec 07, 2024 2:05:21 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ubuntu/jenkins/remoting as a remoting work directory
Dec 07, 2024 2:05:21 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/ubuntu/jenkins/remoting
<===[JENKINS REMOTING CAPACITY]===>channel started
Remoting version: 3261.v9c670a_4748a_9
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```


Build step

Dashboard > master to slave > Configuration

Configure

General

Source Code Management

Build Triggers

Build Environment

Build Steps

Post-build Actions

Terminate a build if it's stuck

With Ant ?

Build Steps

Execute shell ?

Command

See [the list of available environment variables](#)

echo "whatever build in master ends up in slave"

Advanced

Add build step

Save

Apply

 Status

 Changes

 Console Output

 Edit Build Information

 Delete build '#1'

 Timings



Console Output

 Download

 Copy

View as plain text

Started by user [kechiya_jenkins](#)

Running as SYSTEM

Building remotely on [slave-1](#) (dev) in workspace /home/ubuntu/jenkins/workspace/master to slave

[master to slave] \$ /bin/sh -xe /tmp/jenkins9661932662594541632.sh

+ echo whatever build in master ends up in slave

whatever build in master ends up in slave

Finished: SUCCESS

Output:

```
ubuntu@ip-172-31-91-190:~$ cd jenkins
ubuntu@ip-172-31-91-190:~/jenkins$ ls
remoting  remoting.jar  workspace
ubuntu@ip-172-31-91-190:~/jenkins$ cd workspace
ubuntu@ip-172-31-91-190:~/jenkins/workspace$ ls
'master to slave'
ubuntu@ip-172-31-91-190:~/jenkins/workspace$
```